



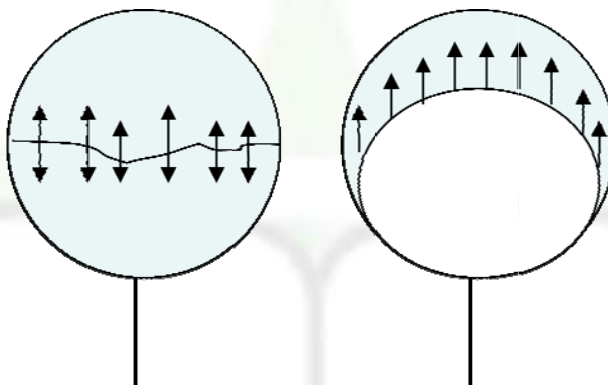
Surface Tension, Angle Of Contact, Capillarity

Surface Tension:

Definition: The tendency of free surface of a liquid to contract can be explained by assuming a force acting parallel to the free surface of the liquid and perpendicular to the boundary at every point. This hypothetical force per unit length of the boundary is known as surface tension.

Unit of Surface Tension: $S = \text{Force} / \text{length}$, Dimension $= [MLT^{-2}]/[L] = [MT^{-2}]$, unit = Nm^{-1}

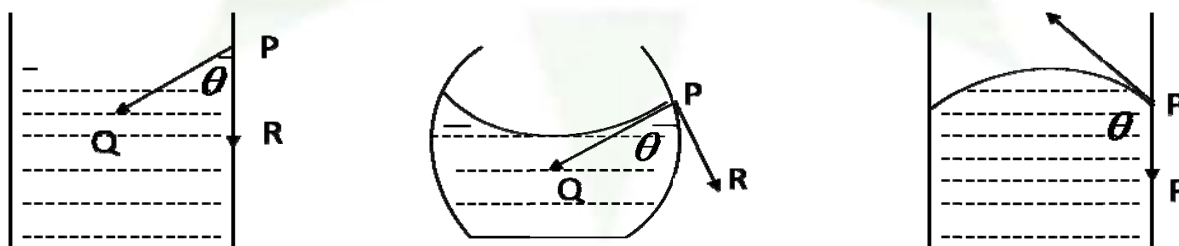
Demonstration: A metal ring provided with a handle with a thread wrapped loosely is taken



along the diameter. The ring is dipped in soap solution and is taken out. A soap film covered the ring. The thread remains loose as it is. Break the lower half of the film, say preaching with a needle. It is found that the string gets taut and is pulled up. This can be explained as due to Surface Tension.

Angle of contact: The angle of contact between a given liquid and solid surface can be defined as the angle between two tangent planes drawn at any point of contact, one tangential to the liquid surface and the other tangential to the wall of the container.

The angle is measured inside the liquid.



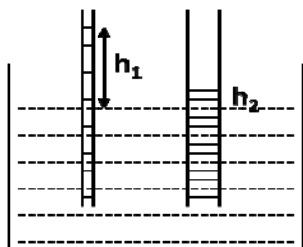
For liquid like water alcohol in contact with glass the cohesive force is very weak compared to the adhesive force hence the shape of the free surface is concave and the angle of contact is acute ($< 90^\circ$).

For liquid like mercury in contact with glass the cohesive force is much stronger than adhesive force the shape of the free surface is convex and the angle of contact is obtuse ($> 90^\circ$).

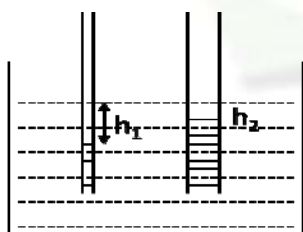


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Capillarity:



Water is taken in a beaker. Capillary tubes of different diameters are dipped into water. It is found that the level of water inside the capillary tube is higher than that in the container i.e. the liquid raises by itself inside the capillary tube.



If the capillary tubes are dipped in mercury the level of mercury falls below the level of Hg in the container.

This phenomenon is known as capillarity.

The height up to which the level rises or falls in the capillary tube is inversely proportional to the radius of the capillary tube.

$$h \propto \frac{1}{r}$$

Jurin's law.



PQ = wall of the container

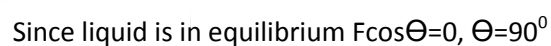
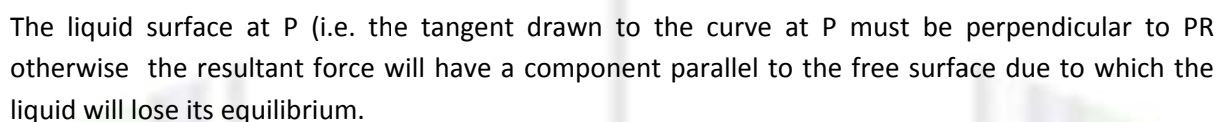
PAB = Free surface of the liquid.

Let us consider a liquid molecule at P on the free surface in contact with the wall of the container.

Force acting on the molecule are :

- (1) Gravitational force of attraction along PN
- (2) Resultant cohesive force along PN_1
- (3) Resultant adhesive force along PN_2

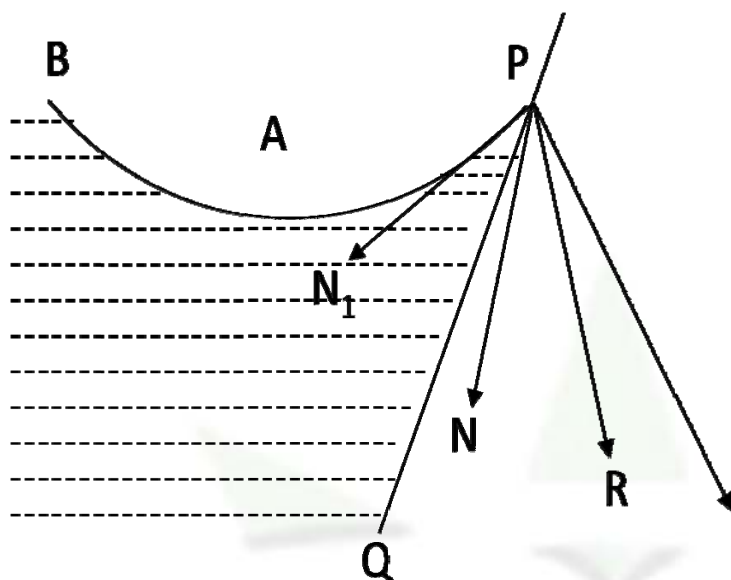
Let $\mathbf{PR}$ be the resultant of these three forces.



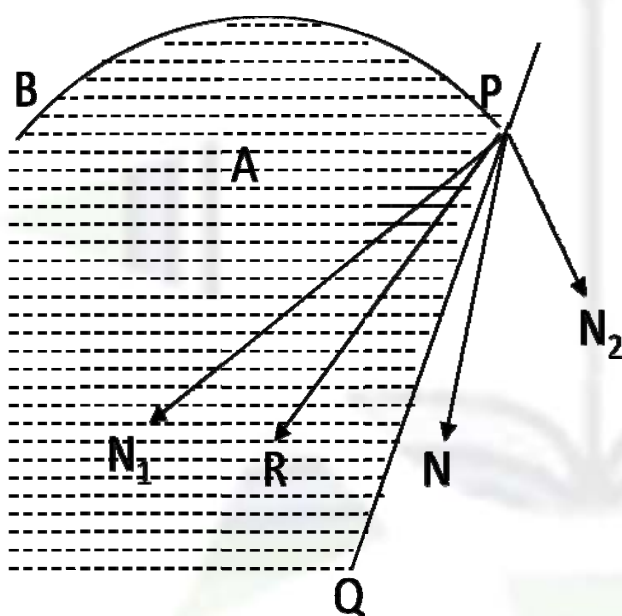
In general PR is not vertical hence the liquid surface at P is not horizontal.



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Case I : For liquid like water alcohol in contact with glass the cohesive force is much weaker than adhesive force. $PN_1 \ll PN_2$. Hence resultant PR lies close to the greater force PN_2 . Hence from the diagram it is clear that the shape of the free surface should be concave.



Case II: For liquid like mercury in contact with glass cohesive force is very much stronger than the adhesive force. $PN_1 \gg PN_2$ and hence the resultant PR is close to PN_1 . Thus from the diagram it is obvious that the shape of the free surface should be convex.